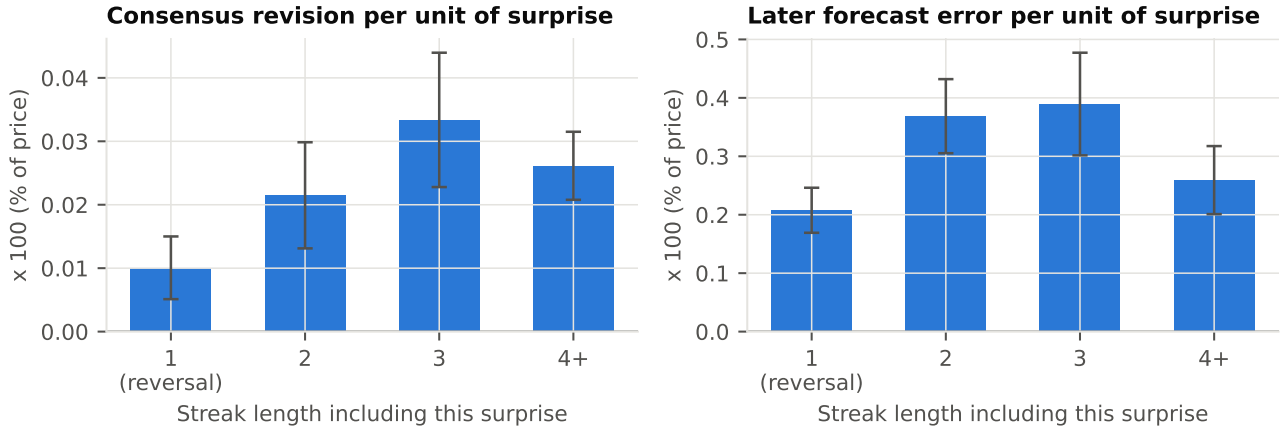
Idea 1 PoC — Do analysts commit the gambler's fallacy?

Proof of concept, October 2026. Code: `idea01_analyst_gamblers_fallacy/run.py`.

Question and design

The gambler's-fallacy paper finds that prices underreact more to earnings surprises that continue a streak. Do sell-side analysts share this belief? Do they revise less after a streak-continuing surprise because they expect the streak to end, and are their forecasts then too conservative? For each quarterly announcement in month m (2005–2020), we measure two things inside I/B/E/S, which shares the price's share basis. The *revision* is the change in the 12-month-forward consensus from the start of m to the start of $m+1$, divided by price. The *forecast error* is the I/B/E/S trailing EPS twelve months later minus the post-announcement consensus, divided by price. A positive slope of the error on the surprise means the consensus still underreacts. Streaks follow the paper: consecutive SRW surprises of the same sign. A surprise with streak length ≥ 2 continues a streak; length 1 is a reversal. All regressions have month fixed effects, with SEs clustered by month, and control for expected growth (forward minus trailing EPS), which absorbs the mechanical drift of the time-weighted consensus. Revisions that span the May fiscal-year roll are dropped. The sample has 34,388 revisions, 26,933 of them with a realised outcome; 65% are continuations.



Slope on SUE (% of price per unit)	Reversal	Extra for continuation	t (extra)
Consensus revision, SRW streaks	0.010	+0.016	4.9
Later forecast error, SRW streaks	0.207	+0.101	4.6
Later forecast error, XGBoost surprises and streaks	0.149	+0.188	8.6
Price drift, months +1.. + 3 (all / covered / not covered)	0.45	+0.13 / +0.17 / +0.00	1.5 / 1.4 / 0.0

Findings

- **Analysts do not expect streaks to reverse.** The consensus moves *more* per unit of surprise after a continuation, and more again with streak length (0.010 for reversals, 0.022–0.033 for streaks). That is the opposite of a naive gambler's fallacy.
- **They still underreact more to streaks, mostly to short ones.** Even after revising, the consensus misses more of a streak surprise. The error slope is 0.21 after reversals, 0.37–0.39 after streaks of 2–3, and only 0.26 after streaks of 4 or more. Underreacting to short streaks and catching up on long ones is what Rabin and Vayanos (2010) predict for believers in the fallacy. The pattern is stronger with XGBoost surprises ($t = 8.6$), which are close to independent, so the streak carries less real information.
- **The link to prices is inconclusive here.** In this simple specification the streak effect in the three-month price drift is positive but insignificant ($t = 1.5$). It is present only in analyst-covered firms (0.17 against 0.00), which fits analysts transmitting the bias, but it is too weak to test mediation.

Assessment and next steps

Promising, and a natural mechanism section or companion paper for the gambler's-fallacy project. Analysts do underweight short streaks, though not by expecting reversals. Next steps: (i) run the price test inside the paper's own Fama–MacBeth framework and add analysts' forecast error as a mediator; (ii) get I/B/E/S detail data (individual analysts and fiscal-year targets) for a clean forecast error and analyst-level fallacy measures, in the spirit of Chen, Moskowitz and Shue (2016); (iii) separate up- and down-streaks, since optimism differs by sign.

References. Rabin and Vayanos (2010, RES); Chen, Moskowitz and Shue (2016, QJE); Loh and Warachka (2012, MS); Abarbanell and Bernard (1992, JF).